

Executive Summary & Technical Brief

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Executive Summary

This document consolidates progress to date on the Hilbert–Huang Transform (HHT) residual analysis of supernova datasets and the construction of a BAO-derived anisotropy sky map.

The analysis focuses on:

- (i) intrinsic mode functions (IMFs) extracted via EMD/EEMD/CEEMDAN and evidence for IMF2/IMF3 “locking”;
 - (ii) the interplay between BAO distance measures — the line-of-sight Hubble distance D_H and the transverse comoving distance D_M — and their ratio $R_{BAO} \equiv D_H/D_M$; and
 - (iii) a new sky-mapping pipeline (SM_SkyMap_FR_KdS.py) that constructs a HEALPix map of R_{BAO} and fits its dipole while comparing to FR/KdS model axes.
- HHT: IMF2 (and sometimes IMF3) shows localized energy ridges consistent with quasi-stationary or gently-chirped content; phase-randomized surrogate tests are used to assess significance.
 - BAO ratio: $R_{BAO} = (D_H/r_d) / (D_M/r_d) = D_H / D_M$ reduces absolute calibration sensitivity and emphasizes geometry.
 - Sky map: Mask-aware HEALPix binning, Gaussian smoothing, dipole fitting with permutation p-values, and redshift tomography implemented.
 - Reproducibility: PNG/FITS/JSON/TXT outputs are written; tomography CSVs summarize per-bin axes and amplitudes.

In parallel, we study two complementary cosmological distance measures: D_H (line-of-sight) and D_M (transverse).

Their ratio R_{BAO} highlights geometry and cancels a common calibration scale.

Projecting R_{BAO} across the sky tests for directional differences (a dipole), and repeating this in redshift slices tracks evolution.

HHT IMF Findings to Date

Methods: EMD, EEMD, and CEEMDAN decompose residuals into IMFs (IMF1, IMF2, IMF3, ...). IMF2 often captures repeatable structure; IMF3 can reflect broader-scale trends.

‘Locking’ refers to alignment between IMF energy ridges and external reference scales.

Robustness: Phase-randomized surrogates (typically 500–1000) and parameter sweeps (pad, grids, seeds) are used to assess stability; outputs are aggregated to CSV/JSON for review.

Observations: Concentrated IMF2 energy appears within specific redshift windows, with hints of gentle chirp-like evolution.

These are candidates for further validation and physical interpretation.

Caveats: HHT is sensitive to endpoints, noise realisations, and method settings; convergence checks across methods/seeds are essential.

BAO Distance Measures and Ratio R_BAO

Key quantities:

$$D_H(z) = c / H(z)$$

$$D_M(z) = (1+z) \cdot D_A(z)$$

$$R_{BAO}(z) = (D_H/r_d) / (D_M/r_d) = D_H / D_M$$

Interpretation: R_BAO compares line-of-sight and transverse scalings at the same redshift.

Because r_d cancels, R_BAO is less sensitive to absolute calibration and more sensitive to geometry.

Coherent deviations can be summarized by a dipole.

R_BAO SkyMap Pipeline (SM_SkyMap_FR_KdS.py)

Input: Unified catalog with RA/Dec or Galactic l/b , redshift z , and BAO columns (DH_{over_rd} , DM_{over_rd}) or (DH , DM , rd).

Optional HHT metrics ($imf2_E$, $imf3_E$, hht_lock) for overlay.

Map: HEALPix binning with mask-aware smoothing (numerator/denominator normalisation) and optional Galactic $|b|$ mask in analysis/display.

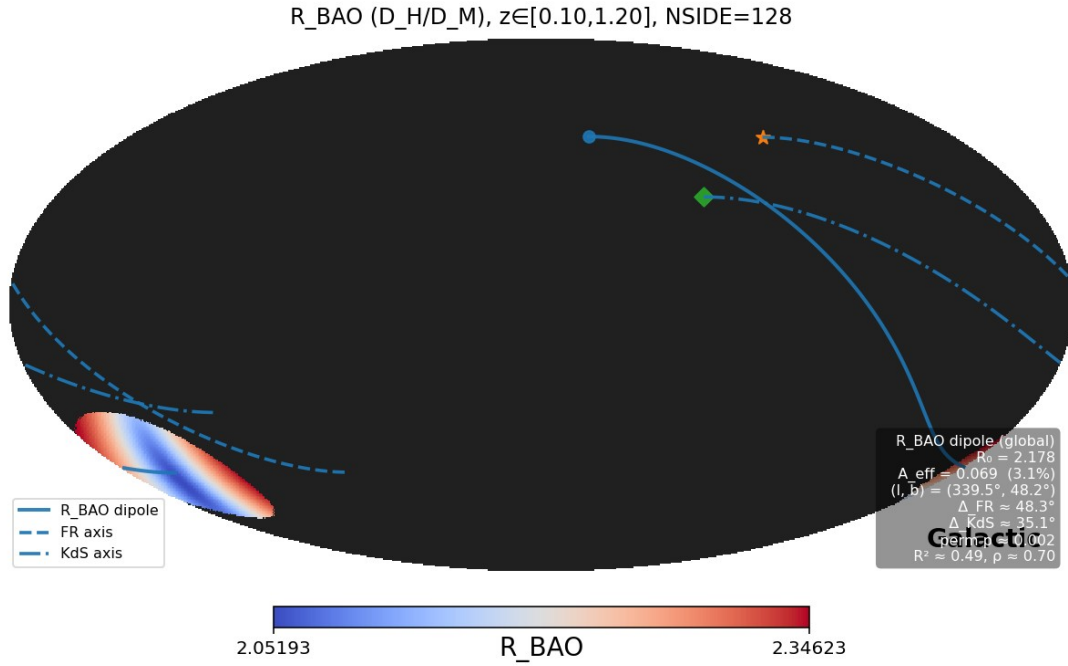
Dipole: Centered linear regression for axis/amplitude with ridge stabilisation; significance from permutations over valid pixels.

Tomography: Repeat mapping in user-defined z -bins; bins with too few valid pixels are skipped.

CSV sidecars capture per-bin results.

Model axes: Overlay and compare measured dipole to FR and KdS axes; report minimum axis separation in degrees.

Skymap HEALPix Map Output



Practical Guidance & Reproducibility

Example Spyder command:

```
%run '/Users/boyde/.spyder-py3/SM_SkyMap_FR_KdS.py' \

'/Users/boyde/.spyder-py3/plamb_runs/SkyMap/data/unified_catalog.csv' \

--out
'/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/SM_RBAO_z0p1_1p2.png' \

--nside 128 \
--zmin 0.1 \
--zmax 1.2 \
--fwhm-smooth-deg 8.0 \
--coord-in icrs \
--rbao-mode auto \
--gal-mask-abs-b-deg 20.0 \
--highlight-quantile 0.90 \
--lock-min 0.60 \
--fr-vec-lb-deg '264,48' \
--kds-vec-lb-deg '300,30' \
--dipole-search-nside 128 \
--cmap coolwarm \
--inset-loc 'lower right' \
--save-stats \
--save-json
```

HHT-BAO-SkyMap Findings (Version 0V3)

Console Out:

```
[analysis] global: valid_pix = 4443/196608 (2.3%)

[dipole-fit  $\perp$  1] R0 = 2.17777 A_eff = 0.0685007 (frac = 3.145% ) cond  $\approx$  7.58e+01

[hemi-contrast]  $\Delta$  = 0.100039 (frac = 4.594% )

[null|permute] n=500 p(A_null  $\geq$  A_obs)  $\approx$  0.0020

[angle ] dipole-FR = 48.27 deg (axis-min = 48.27 deg)

[angle ] dipole-KdS = 35.12 deg (axis-min = 35.12 deg)

[dipole-dir] l = 339.53 b = 48.22

[fit-quality] R2  $\approx$  0.485 ( $\approx$  corr2), corr(y,cos $\theta$ _axis)  $\approx$  0.697

[amplitude] A_sigma  $\approx$  0.060355 (2.771% of mean), A_range  $\approx$  0.122090 (5.606% of mean)

[Y1-check] |a_l=1|  $\approx$  0.295202 (mask-biased sanity check)

[X-power] no template_map provided; skipping cross-power.

[analysis] z $\in$ [0.10,0.40]: valid_pix = 4434/196608 (2.3%)

[tomography] z $\in$ [0.10,0.40]: l=158.77 b_deg=-48.36 R0=2.42322 A_eff=5.05034e-14 (frac=0.00%)

[analysis] z $\in$ [0.40,0.70]: valid_pix = 4001/196608 (2.0%)

[tomography] z $\in$ [0.40,0.70]: l=48.41 b_deg=13.73 R0=1.62465 A_eff=0.140957 (frac=8.68%)

[analysis] z $\in$ [0.70,1.00]: valid_pix = 3592/196608 (1.8%)

[tomography] z $\in$ [0.70,1.00]: l=52.88 b_deg=27.20 R0=0.981772 A_eff=0.0509945 (frac=5.19%)

[analysis] z $\in$ [1.00,1.20]: valid_pix = 2740/196608 (1.4%)

[tomography] z $\in$ [1.00,1.20]: l=12.44 b_deg=58.87 R0=0.823463 A_eff=1.81724e-15 (frac=0.00%)

[tomography] saved  $\rightarrow$ 
/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/SM_RBAO_z0p1_1p2_tomography.csv

[X-power] C1(map, FR)  $\approx$  -0.131126

[X-power] C1(map, KdS)  $\approx$  -0.164876

setting the output map dtype to [dtype('float64')]

[ok] saved stats  $\rightarrow$ 
/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/SM_RBAO_z0p1_1p2_stats.txt
```

HHT-BAO-SkyMap Findings (Version 0V3)

<Figure size 864x576 with 0 Axes>

Outputs:

- PNG sky map,
- FITS map,
- JSON summary (dipole, tomography),
- and optional TXT stats;
- tomography CSVs saved alongside the PNG.

Acronyms

Acronym	Meaning
BAO	Baryon Acoustic Oscillation
HHT	Hilbert–Huang Transform
IMF	Intrinsic Mode Function
EMD/EEMD/CEEMDAN	Empirical Mode Decomposition / Ensemble EMD / Complete Ensemble EMD with Adaptive Noise
HEALPix	Hierarchical Equal Area isoLatitude Pixelization
FR	Full Relativity (model axis)
KdS	Kerr–de Sitter (model axis)
R_BAO	Distance ratio D_H/D_M
NSIDE	HEALPix resolution parameter

Variables & Symbols

Symbol	Description	Units
C	Speed of light	km s^{-1}
H(z)	Hubble expansion rate at redshift z	$\text{km s}^{-1} \text{Mpc}^{-1}$
D_H(z)	Hubble distance $c/H(z)$	Mpc
D_A(z)	Angular diameter distance	Mpc
D_M(z)	Transverse comoving distance $= (1+z) \cdot D_A$	Mpc
r_d	Sound horizon at the drag epoch	Mpc
R_BAO(z)	BAO distance ratio D_H/D_M	Dimensionless
Z	Redshift	—
l, b	Galactic longitude/latitude	Deg
NSIDE	HEALPix map resolution parameter	—
imf2_E, imf3_E	IMF2/IMF3 energy metric	arb. Units
hht_lock	HHT locking score or filter	0–1 (typ.)

Limitations & Next Steps

- Validate stability across HHT methods and seeds; record effect sizes and false-positive rates.
- Use inverse-variance weights whenever available (σ columns).
- Check systematics: Galactic masking thresholds, smoothing scales, and catalog sub-selections.
- Extend tomography with adaptive binning to maintain pixel counts.
- Cross-validate with independent datasets or mocks.